INCH-ROUND

MIL-M-28787/302A
AMENDMENT 4
30 July 1993
SUPERSEDING
AMENDMENT 3
11 March 1985

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

MEMORY, RANDOM ACCESS, KEY CODE ZJS

This amendment forms a part of Military Specification MIL-M-28787/302A(EC), dated 10 July 1981, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 2

Add the following new paragraphs:

"3.5.8.1 Test code marking. A test code in accordance with 3.5.8.2 shall be marked on the module in the area specified for manufacturer's information in accordance with MIL-STD-1389. This test code is to identify the vendor's integrated circuit used on a module. The identity of the integrated circuit vendor is indicated to facilitate topology dependent testing; therefore, the module shall be tested with the topology corresponding to the integrated circuit vendor used. A test code shall be assigned by the SEM-QAA for each vendor's integrated circuit type used. Additional test code identification numbers shall be requested from and assigned by the SEM-QAA. Test code identification numbers may be marked on the modules upon assignment."

"3.5.8.2 Test code identification numbers. The following test code identification numbers have been assigned.

Test code identification number	Integrated circuit manufacturer's FSCM
00	50088 (Mostek)
01	04713 (Motorola)
02	34335 (Advanced Micro Devices)
03	01295 (Texas Instruments)

An example of the test code identification is as follows: TEST IDENT - 00"

"3.5.8.3 Test code exceptions. Modules built to this detail specification prior to MIL-M-28787/302A(EC) Amendment 3 were not required to have test code marking."

"3.10 Testability requirements. Dynamic random access memory integrated circuits of different vendors or designs shall not be mixed on a module. In addition, address scrambling between the integrated circuit and the module level shall be avoided."

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PAGE 4

*TABLE I, Test condition D, Test type column: Delete "X" and substitute "S".

TABLE I, Test condition column: Delete "J1-J5" and substitute "J1-J4"; delete "J6-J10" and substitute "J5-J8".

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*4.4.2: Delete and substitute the following:

"4.4.2 Order of testing. Test conditions A through X shall be performed in any order with the following exceptions. Test conditions A, B, C, and D are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, B, C, or D are part of a sequence of tests, the order of testing shall be as follows: A, B, C, and D where applicable, followed by the remaining tests in the sequence."

*4.5.1, line 5: Delete "D through V" and substitute "E through V".

*4.5.3, line 3: Delete "D through V" and substitute "E through V".

*4.5.3, line 6: Delete "D through X" and substitute "E through X".

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FIGURE 6: Add a connection dot at the intersection of the E line from the negative terminal of item P01-3 and the 0 V REF line.

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FIGURE 7: Add the following on figure 7:

" CAUTION:
Do not isolate the chassis of
ammeter X01 from earth ground."

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TABLE VIII: Delete and substitute new table VIII as printed on page 4 of this amendment.

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FIGURE 8: Delete and substitute new figure 8 as printed on page 5 of this amendment.

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FIGURE 11: Delete the note in its entirety.

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FIGURE 12: Delete and substitute new figure 12 as printed on page 6 of this amendment.

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FIGURE 18: Delete and substitute new figure 18 as printed on page 7 of this amendment.

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*FIGURE 41, ITEM C01 box: Delete "C01" and substitute "C06".

*FIGURE 41, NOTE 3, first line: Delete "(C01)" and substitute "(C06)".

FIGURE 41, NOTE 3, second line: Delete "Frequency = 120 Hz" and substitute "Frequency = 1 KHz."

NOTE: The margins of this amendment are marked with an asterisk or vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

Custodians:
Army - ER
Navy - NW
Air Force - 85

Preparing activity:
Navy - NW

(Project 5963-0234)

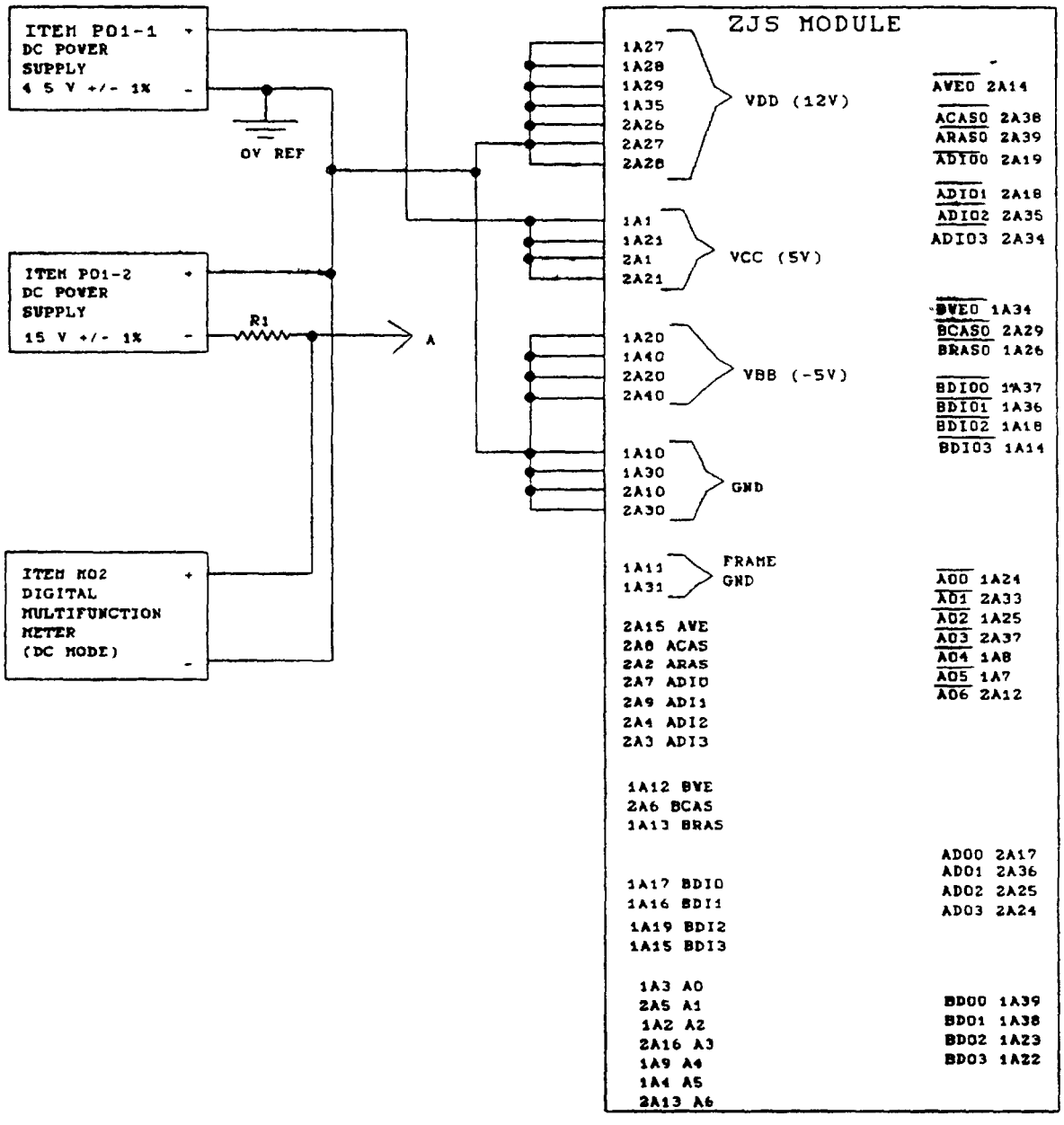
Review activities:
Army - AT
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Air Force - 13, 19
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TABLE VIII. Ground pin continuity test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
J1	1A30 Ground pin cont.	1A30	1A10	0	1	Ω	11
J2	2A10	2A10	1A10				
J3	2A30	2A30	1A10				
J4	Frame	1A11	1A31	0	1		
J5	1A30	1A30	1A10	0.0	0.1		
J6	2A10	2A10	1A10				
J7	2A30	2A30	1A10				
J8	Frame Ground pin cont.	1A11	1A31	0.0	0.1	Ω	11

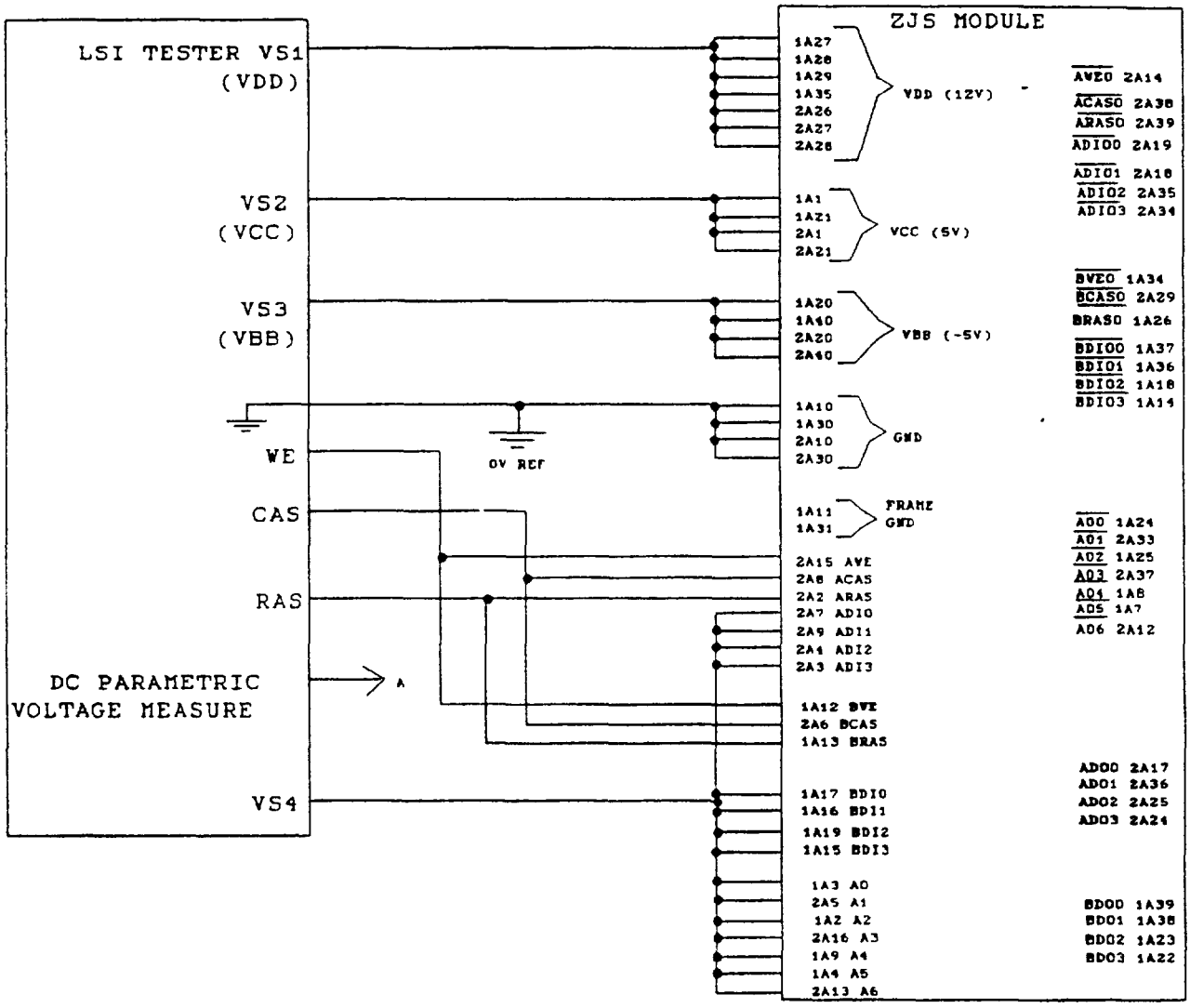
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NOTE: R1 = 750Ω ± 1%, 1/8 W.

FIGURE 8. Input clamp diode voltage test circuit for table V.

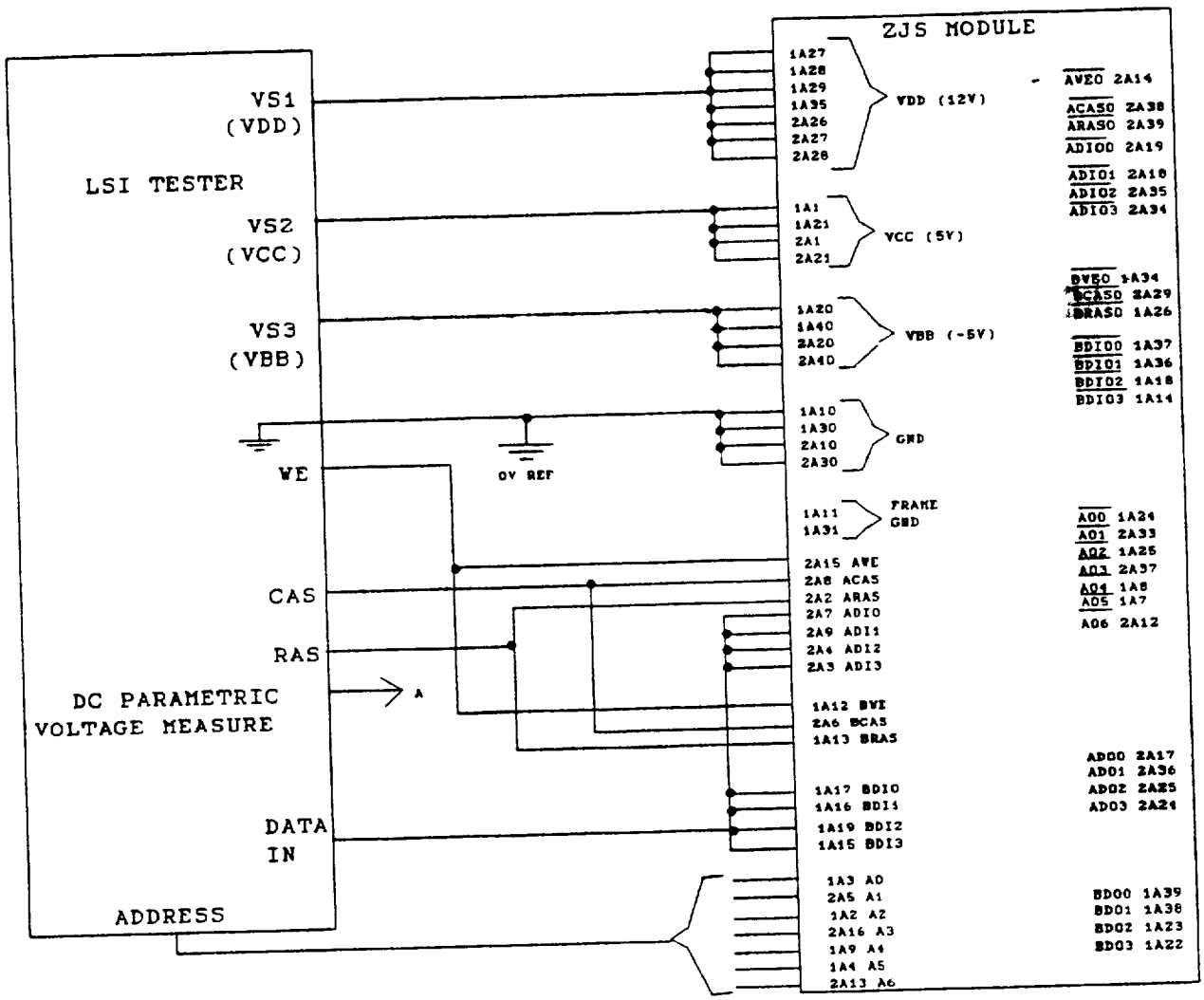
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NOTES:
1. All tester leads to be of minimum length.
2. Input conditions are found in table X for test conditions K1-K8 and table XI for test conditions L1-L8.
3. Load outputs per figure 14.

FIGURE 12. Output voltage (low-level and high-level) test circuit for table IX.

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NOTES:
1. **All tester leads to be 2 inches (50.8 mm) maximum length.**
2. **Load each output per figure 14.**

FIGURE 18. Dynamic functional test circuit for table XIII.

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